

FuelPrice

Class: FuelPrice - Front	
Superclass: none Sub-class: none	
Responsibilities	Collaborators
Know its amount, submission time and lifecycle status	-
Validate the submitted amount against the locked daily cap	DailyPriceCap
Activate a validated price so it is shown to motorists	FuelStation
Expire / supersede itself when a newer price is activated or it goes stale	FuelStation, Scheduler (flag stale)
Know which fuel type it prices	FuelType
Supply price data to search results	SearchRequest

Class: FuelPrice - Back
Attributes - priceId : String - amount : Float - submittedAt : DateTime - status : String (submitted / validating / active / rejected / stale)
Relationships Generalisation: none Association: - FuelType (0..* prices to 1 fuelType); - DailyPriceCap (0..* checkedPrices to 1 appliedCap) Aggregation: none Composition: part of FuelStation (station 1 to 0..* priceHistory)
Business Rules - A submitted amount must not exceed the locked DailyPriceCap for that fuel type and effective date. - Only a validated price can become active; activating a new price supersedes the previous active price for that fuel type. - A price not refreshed by the retailer is flagged stale by the Scheduler and removed from display.

PriceAlert

Class: PriceAlert - Front	
Superclass: none Sub-class: none	
Responsibilities	Collaborators
Know fuel type, threshold price, radius and quiet hours	-
Belong to the account holder who created it	RegisteredAccountHolder
Evaluate newly activated prices against threshold and radius	FuelPrice, FuelStation, Location
Trigger and own notifications when a match occurs, respecting quiet hours	AlertNotification
Pause / resume / delete on the holder's request	RegisteredAccountHolder
Know its lifecycle status (active / paused)	-

Class: PriceAlert - back
Attributes - alertId : String - fuelCode : String - thresholdPrice : Float - radiusKm : Float - quietHours : TimeRange - status : String (active / paused)
Relationships Generalisation: none Association: RegisteredAccountHolder (0..* savedAlerts to 1 holder); FuelPrice (0..* matchedAlerts to 0..* triggeringPrices) Aggregation: none Composition: whole of AlertNotification (1 to 0..*) - notifications cannot exist without their alert
Business Rules - Only a RegisteredAccountHolder can create an alert - guest users must register first (integration with generalisation). - An alert triggers only when an Active FuelPrice for its fuelCode, within radiusKm, falls to or below thresholdPrice. - No notifications during quietHours - matches are deferred and sent once quiet hours end.

AlertNotification

Class: AlertNotification - Front

Superclass: none

Sub-class: none

Responsibilities	Collaborators
Know its channel, send time and delivery status	-
Compose the alert message (station, fuel, new price, saving)	PriceAlert, FuelPrice, FuelStation
Deliver the message through the chosen channel (push / email)	Notification Service (push provider)
Confirm delivery, or retry when delivery fails	PriceAlert
Record the triggering price so the holder can verify the saving	FuelPrice

Class: AlertNotification - Back

Attributes

- notificationId : String
- channel : String (push / email)
- sentAt : DateTime
- deliveryStatus : String (pending / delivered / failed)

Relationships

Generalisation: none

Association: none (reaches FuelPrice and the holder via its owning PriceAlert)

Aggregation: none

Composition: part of PriceAlert (alert 1..1 to 0..* notifications)

Business Rules

- A notification is created only by a triggered PriceAlert - never directly by a user or another class.
- At most one notification per alert per price-drop event (no duplicate alerts for the same price activation).
- Failed deliveries are retried up to 3 times; after that deliveryStatus = failed and the holder is notified in-app.